X-Ray Diffraction

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- How Diffraction Works
  - Demonstration
  - Analyzing Diffraction Patterns
- Solving DNA
- Applications
- Summary and Conclusions

Introduction

Motivation:

- X-ray diffraction is used to obtain structural information about crystalline solids.
- Useful in biochemistry to solve the 3D structures of complex biomolecules.
- Bridge the gaps between physics, chemistry, and biology.

X-ray diffraction is important for:

- Solid-state physics
- Biophysics
- Medical physics
- Chemistry and Biochemistry

X-ray Diffractometer

History of X-Ray Diffraction

1895 X-rays discovered by Roentgen

1914 First diffraction pattern of a crystal made by Knipping and von Laue

1915 Theory to determine crystal structure from diffraction pattern developed by Bragg.

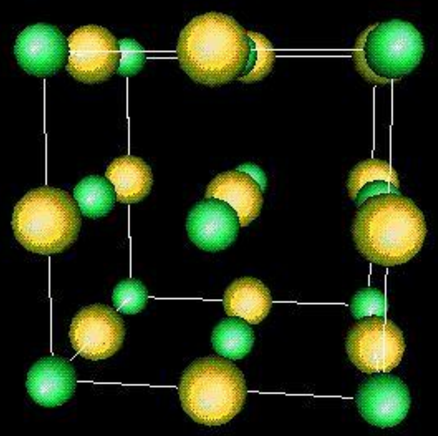
1953 DNA structure solved by Watson and Crick

Now Diffraction improved by computer technology; methods used to determine atomic structures and in medical applications

The first X-ray

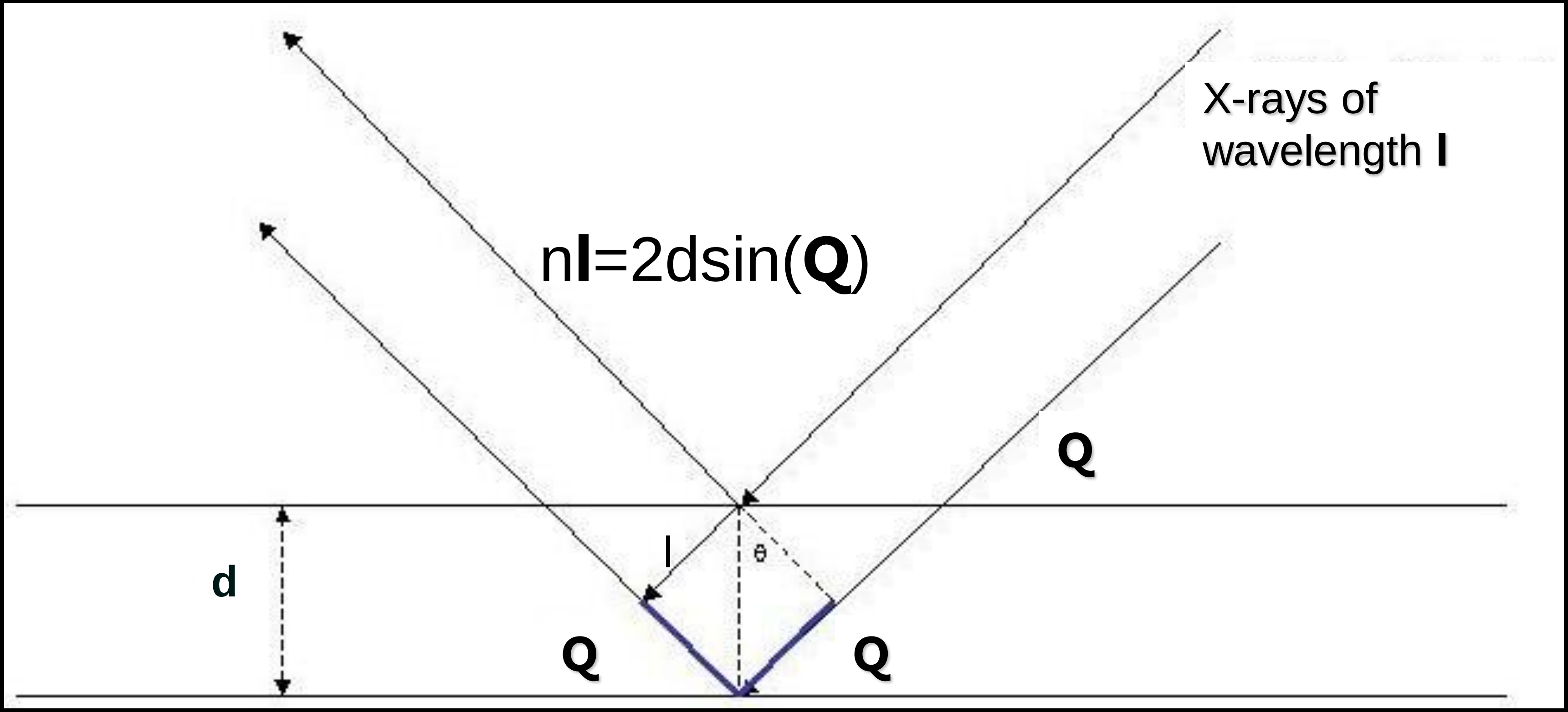
How Diffraction Works

- Wave Interacting with a Single Particle
  - Incident beams scattered uniformly in all directions
- Wave Interacting with a Solid
  - Scattered beams interfere constructively in some directions, producing diffracted beams
  - Random arrangements cause beams to randomly interfere and no distinctive pattern is produced
- Crystalline Material
  - Regular pattern of crystalline atoms produces regular diffraction pattern.
  - Diffraction pattern gives information on crystal structure



NaCl

How Diffraction Works: Bragg's Law



- Similar principle to multiple slit experiments
- Constructive and destructive interference patterns depend on lattice spacing (d) and wavelength of radiation (l)
- By varying wavelength and observing diffraction patterns, information about lattice spacing is obtained

How Diffraction Works: Schematic

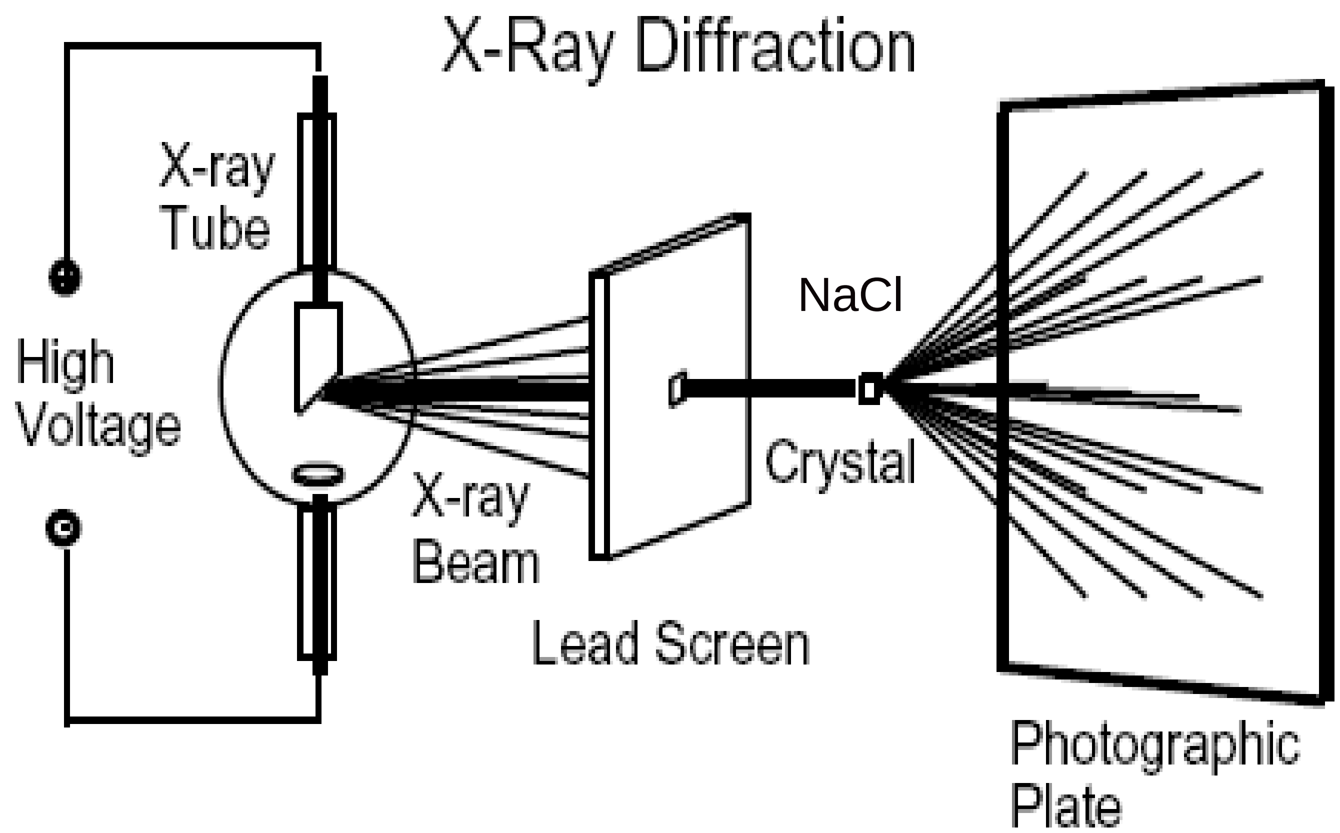


Figure 2. A schematic of X-ray diffraction.

How Diffraction Works: Schematic

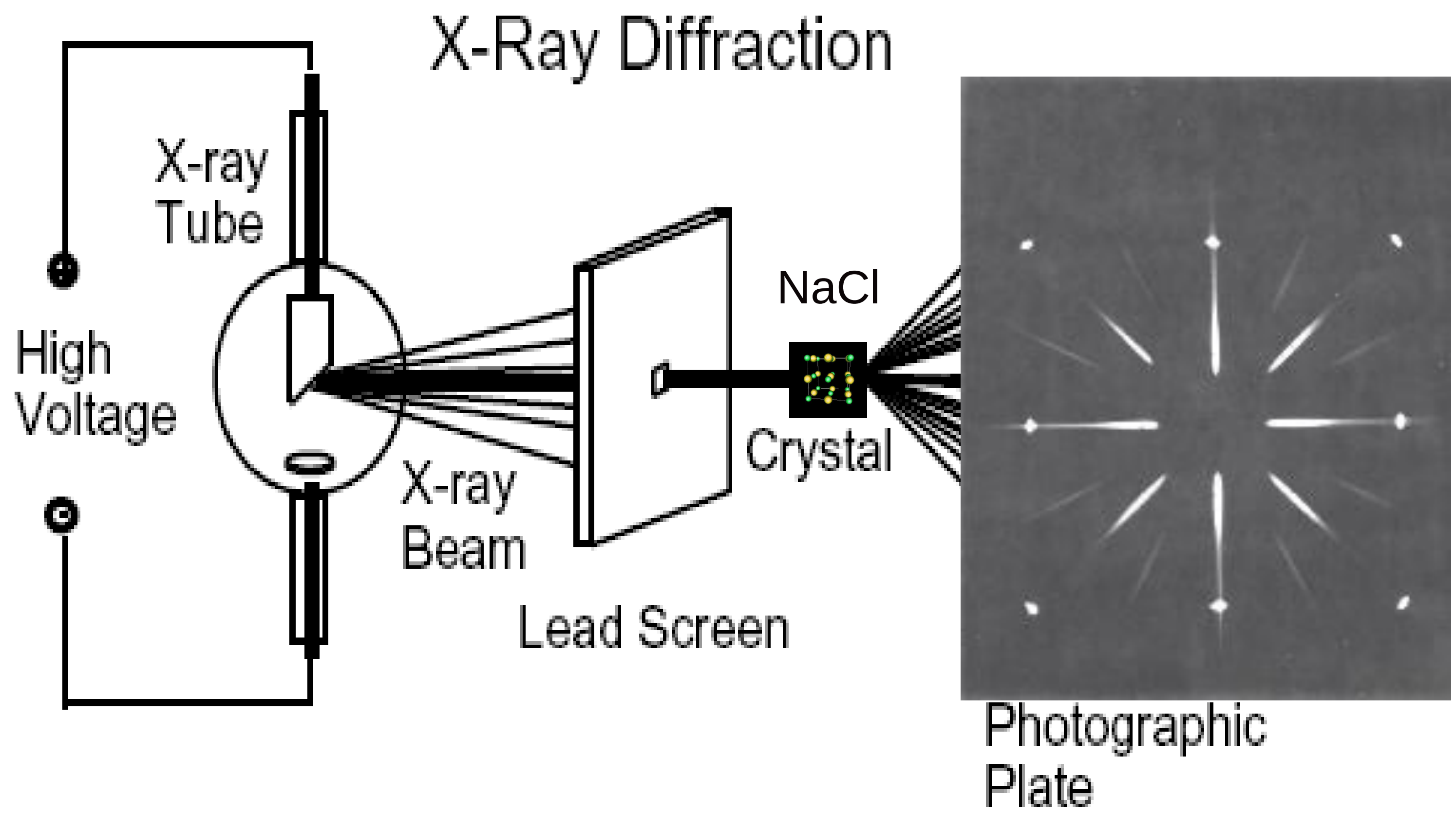


Figure 2. A schematic of X-ray diffraction.

http://mrsec.wisc.edu/edetc/modules/xray/X-raystm.pdf

Demonstration

Array A versus Array B

- Dots in A are closer together than in B
- Diffraction pattern A has spots farther apart than pattern B

Array E

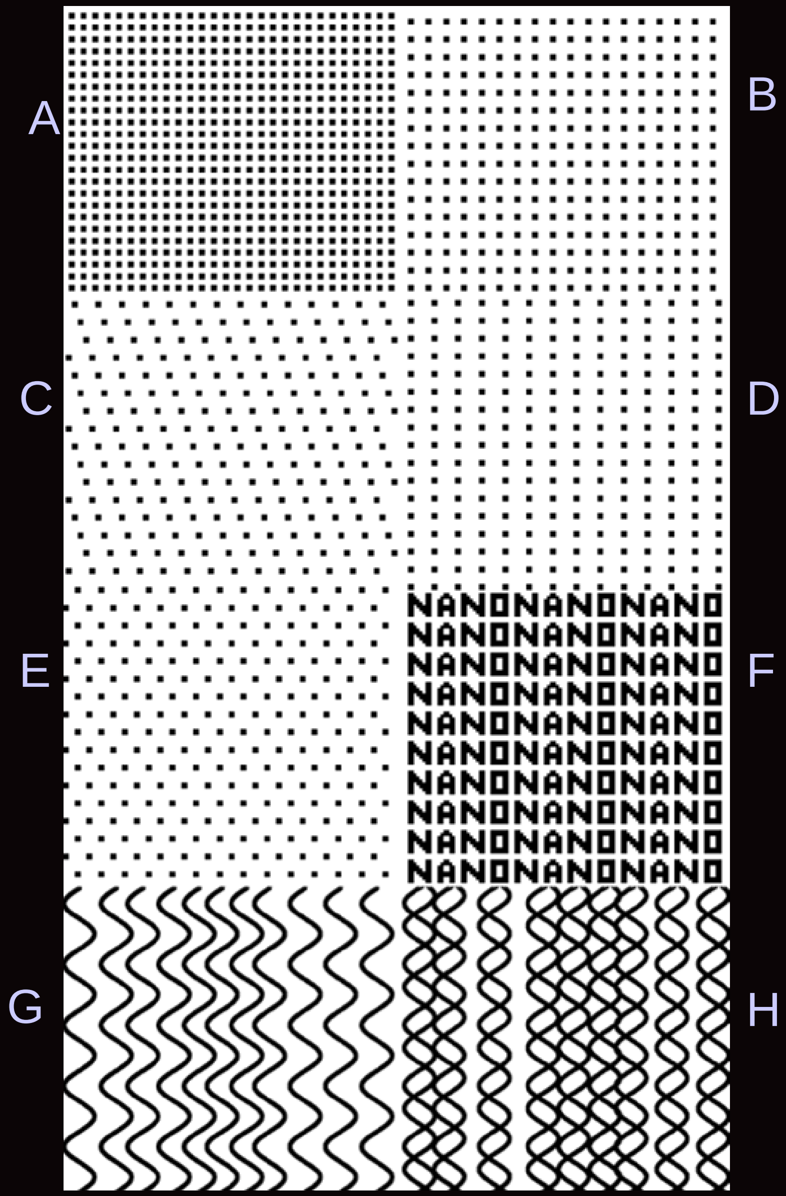
- Hexagonal arrangement

Array F

- Pattern created from the word “NANO” written repeatedly
- Any repeating arrangement produces a characteristic diffraction pattern

Array G versus Array H

- G represents one line of the chains of atoms of DNA (a single helix)
- H represents a double helix
- Distinct patterns for single and double helices

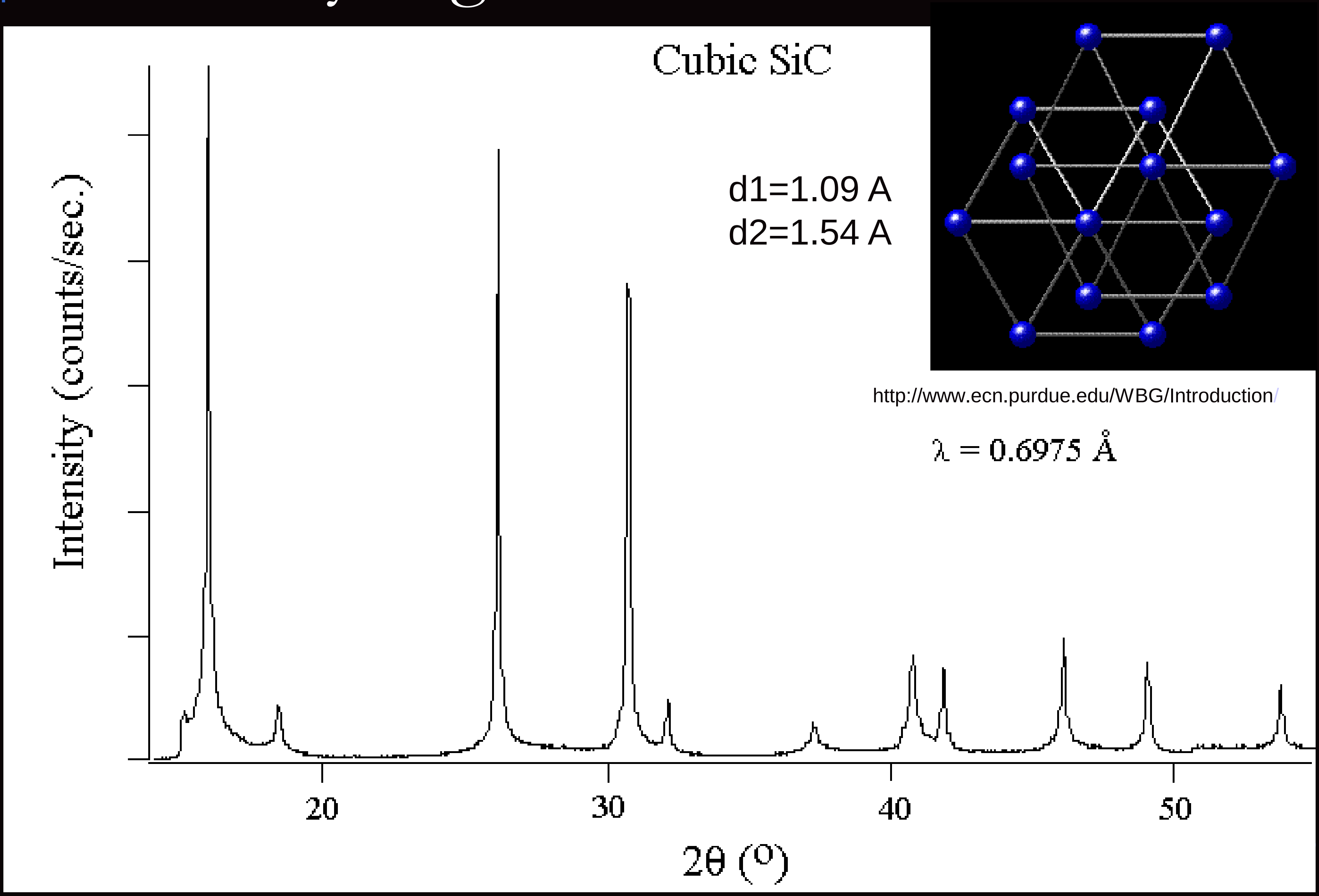


Credit: Exploring the Nanoworld

Analyzing Diffraction Patterns

- Data is taken from a full range of angles
- For simple crystal structures, diffraction patterns are easily recognizable
- Phase Problem
  - Only intensities of diffracted beams are measured
  - Phase info is lost and must be inferred from data
- For complicated structures, diffraction patterns at each angle can be used to produce a 3-D electron density map

Analyzing Diffraction Patterns



d1=1.09 A
d2=1.54 A

http://www.ecn.purdue.edu/WBG/Introduction/

$\lambda = 0.6975$ Å

http://www.eserc.stonybrook.edu/ProjectJava/Bragg/

nl=2dsin(Q)

Solving the Structure of DNA: History

- Rosalind Franklin- physical chemist and x-ray crystallographer who first crystallized and photographed BDNA
- Maurice Wilkins- collaborator of Franklin
- Watson & Crick- chemists who combined the information from Photo 51 with molecular modeling to solve the structure of DNA in 1953

Rosalind Franklin

Solving the Structure of DNA

- Photo 51 Analysis
  - “X” pattern characteristic of helix
  - Diamond shapes indicate long, extended molecules
  - Smear spacing reveals distance between repeating structures
  - Missing smears indicate interference from second helix

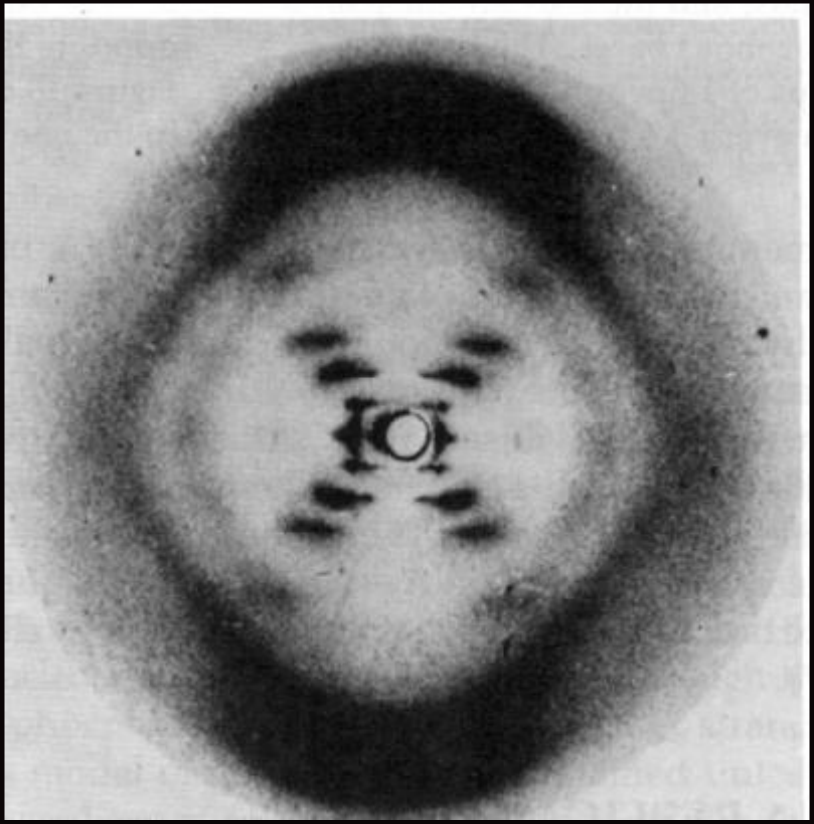


Photo 51- The x-ray diffraction image that allowed Watson and Crick to solve the structure of DNA

www.pbs.org/wgbh/nova/photo51

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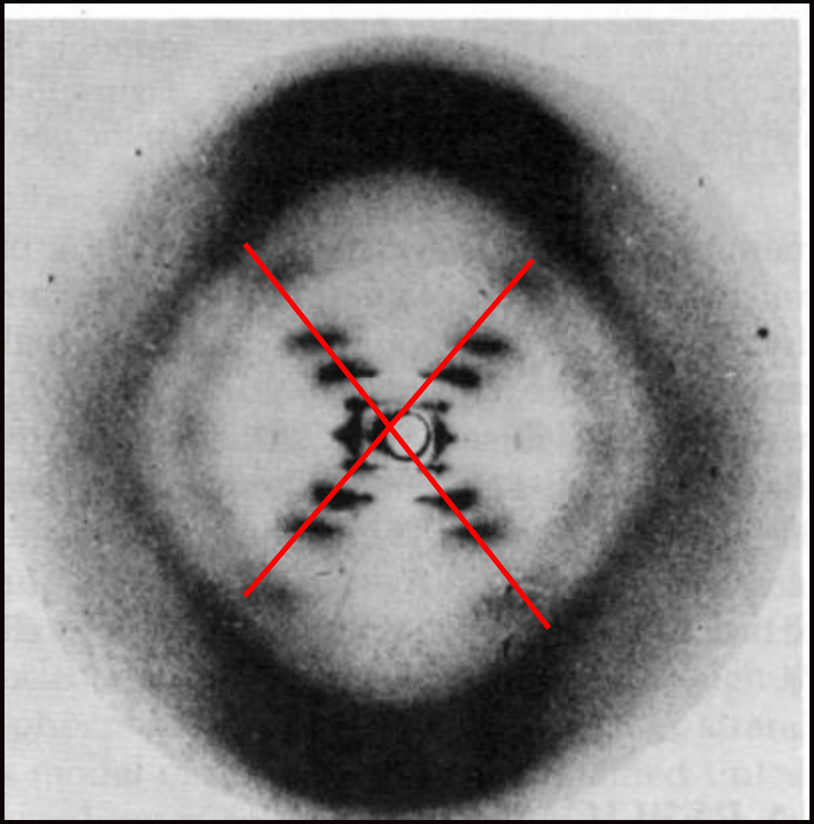

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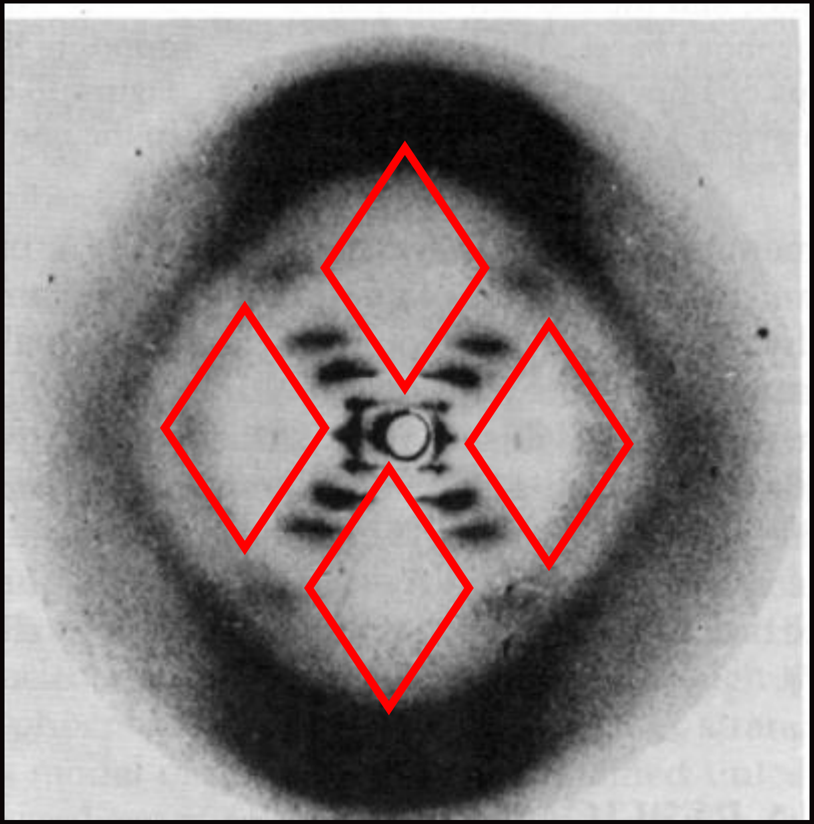

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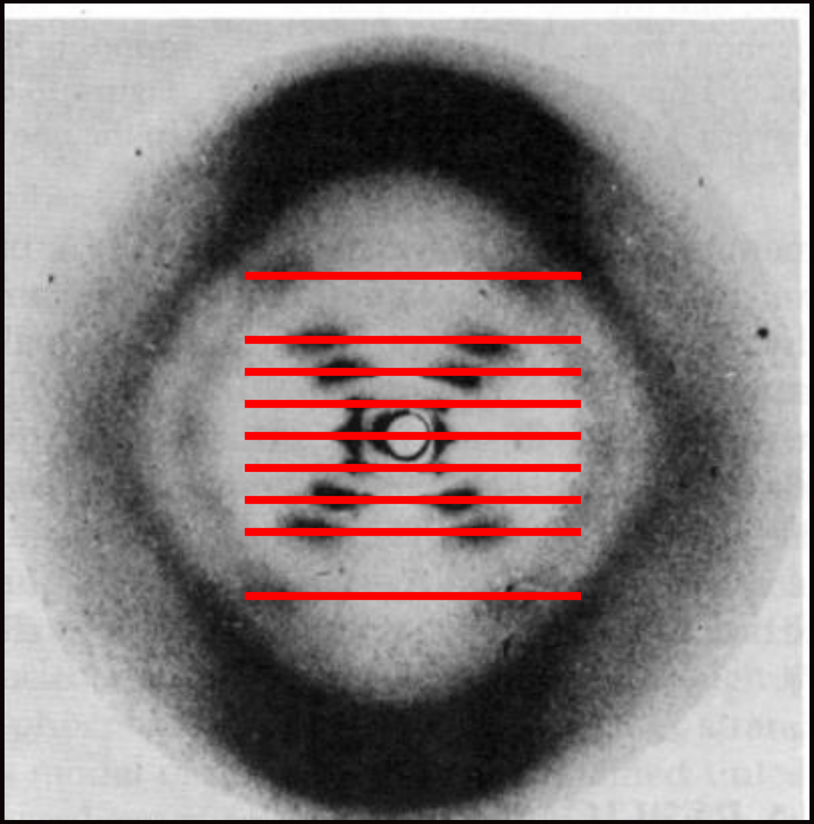

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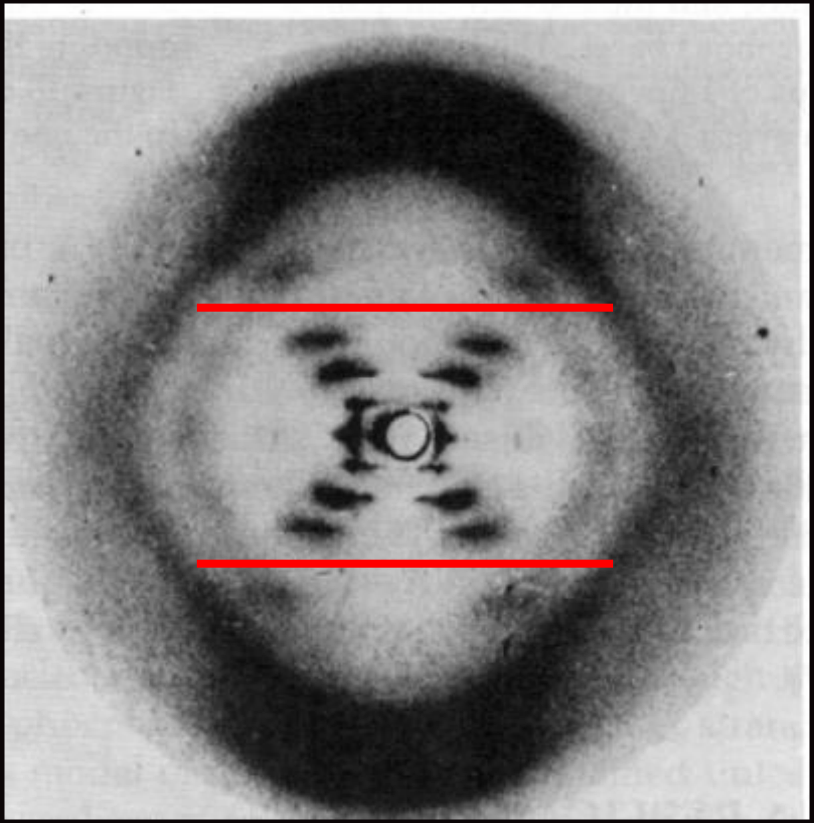
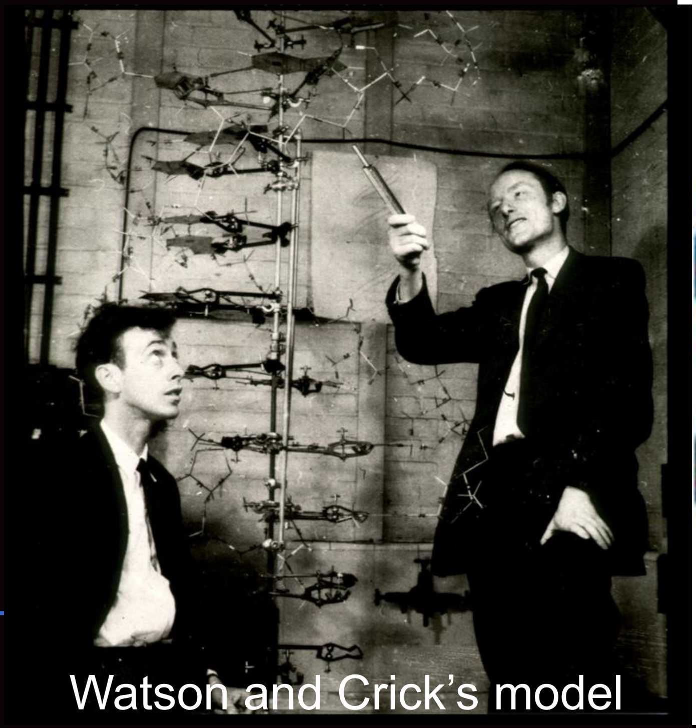


Photo 51- The x-ray diffraction image that allowed Watson and Crick to solve the structure of DNA

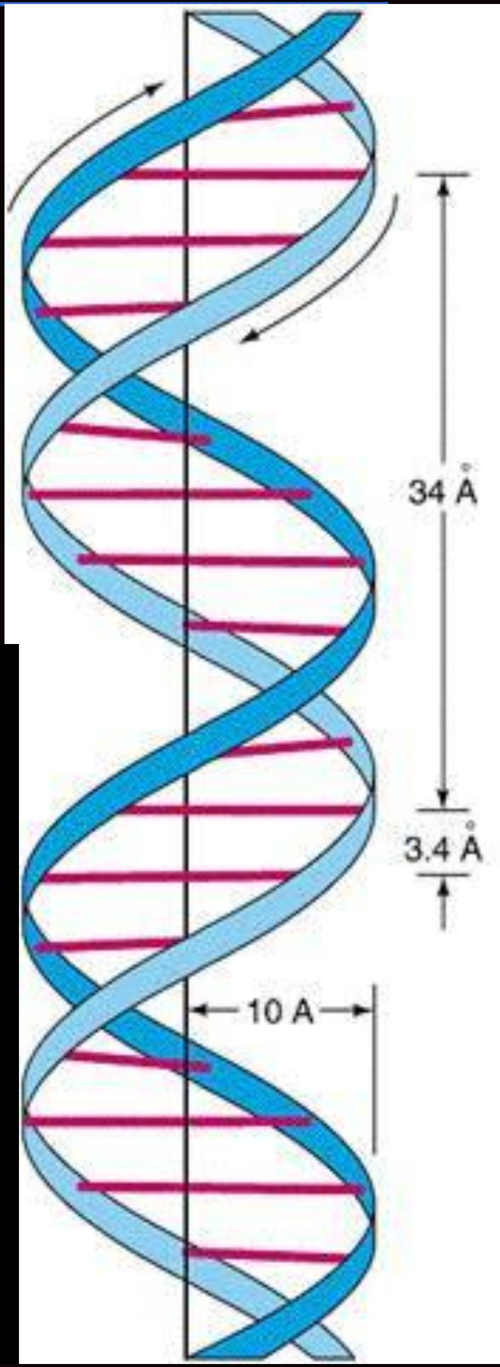
www.pbs.org/wgbh/nova/photo51

Solving the Structure of DNA

- Information Gained from Photo 51
  - Double Helix
  - Radius: 10 angstroms
  - Distance between bases: 3.4 angstroms
  - Distance per turn: 34 angstroms
- Combining Data with Other Information
  - DNA made from:
    - sugar
    - phosphates
    - 4 nucleotides (A,C,G,T)
  - Chargaff's Rules
    - %A=%T
    - %G=%C
  - Molecular Modeling



Watson and Crick's model



Applications of X-Ray Diffraction

- Find structure to determine function of proteins
- Convenient three letter acronym: XRD
- Distinguish between different crystal structures with identical compositions
- Study crystal deformation and stress properties
- Study of rapid biological and chemical processes
- …and much more!

Summary and Conclusions

- X-ray diffraction is a technique for analyzing structures of biological molecules
- X-ray beam hits a crystal, scattering the beam in a manner characterized by the atomic structure
- Even complex structures can be analyzed by x-ray diffraction, such as DNA and proteins
- This will provide useful in the future for combining knowledge from physics, chemistry, and biology

Questions?

References

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Exploring the Nanoworld
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